Supplementary Table S1.

Species	Stages of development of differential expression analysis	Genomes and differentially expressed genes (DEGs)	Authors
Arabidopsis thaliana	• Stage 1-12 • Stage 1-12 • Inflorescence meristem, stage 1-12 and 9. • 3, 6, 9, 12 days after pollination.	Genome DEGs	• Alvarez-Buylla et al., 2010. • Max-Planck Gesellschaft, 2024. • Zhang et al., 2015 • Mizzotti et al., 2018
Vitis vinifera	• Lateral bud and dormant bud • 3, 5, and 7 days after pollination.	Genome DEGs	• Min et al., 2017 • Domingos et al., 2016
Vanilla planifolia	• 1 day before pollination, anthesis, 25 and 60 days after pollination	Transcriptome DEGs	• Hernándes-Miranda et al., 2025
Solanum lycopersicum	• Flower buds and flowers. • Anthesis and 4 days after flowering	Genome	• Wang et al., 2021 • Pattison et al., 2015

Supplementary Figure S1.

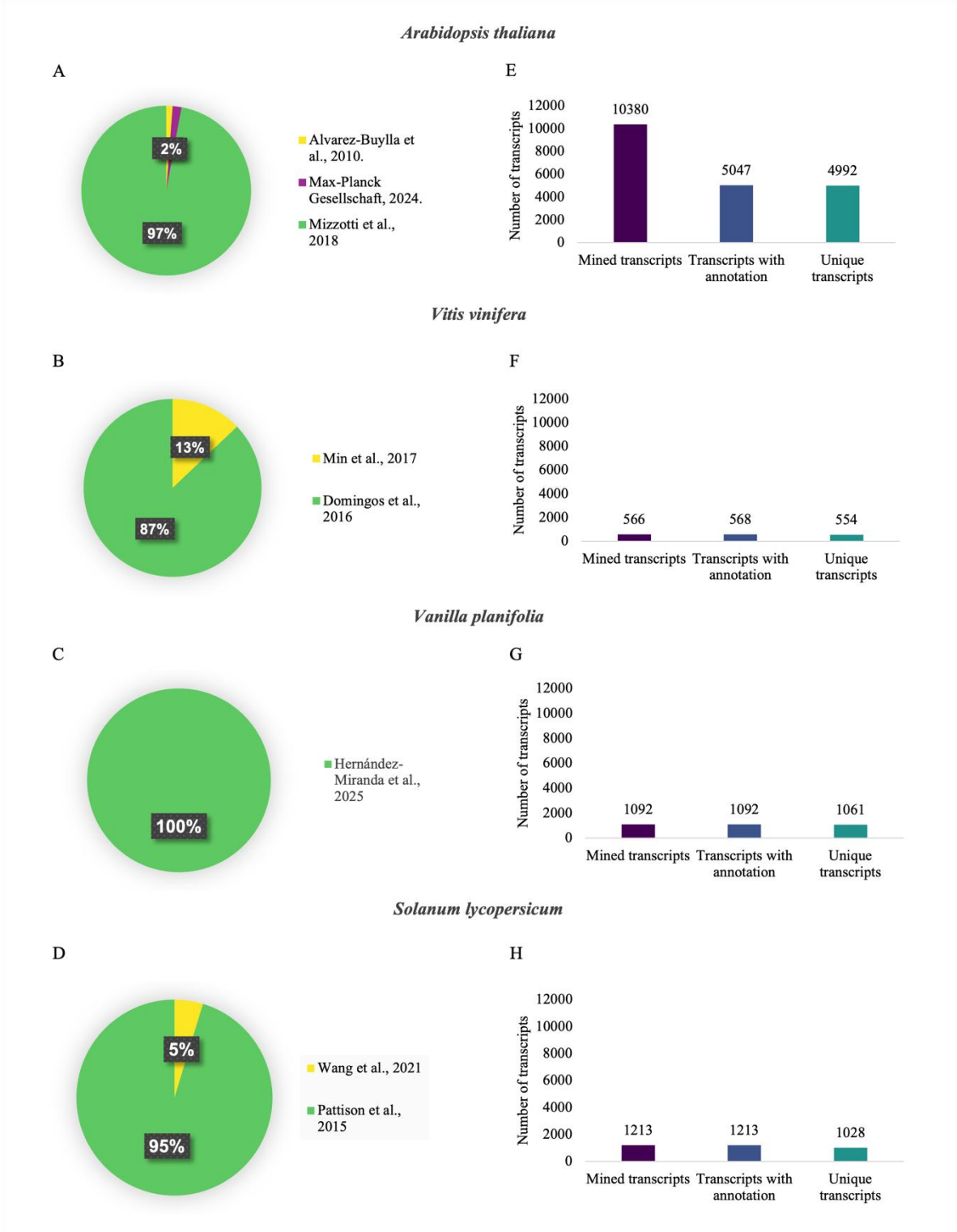


Fig. S1. Distribution of genes associated with flower-to-fruit transition in angiosperms. (A–D) Pie charts showing the proportion of genes retrieved from different databases and previous publications for *Arabidopsis thaliana* (A), *Vitis vinifera* (B), *Vanilla planifolia* (C), and *Solanum*

lycopersicum (D). (E–H) Bar plots showing the total number of mined genes, functionally annotated genes, and unique genes for each species: *A. thaliana* (E), *V. vinifera* (F), *V. planifolia* (G), and *S. lycopersicum* (H).

Supplementary Figure S2.

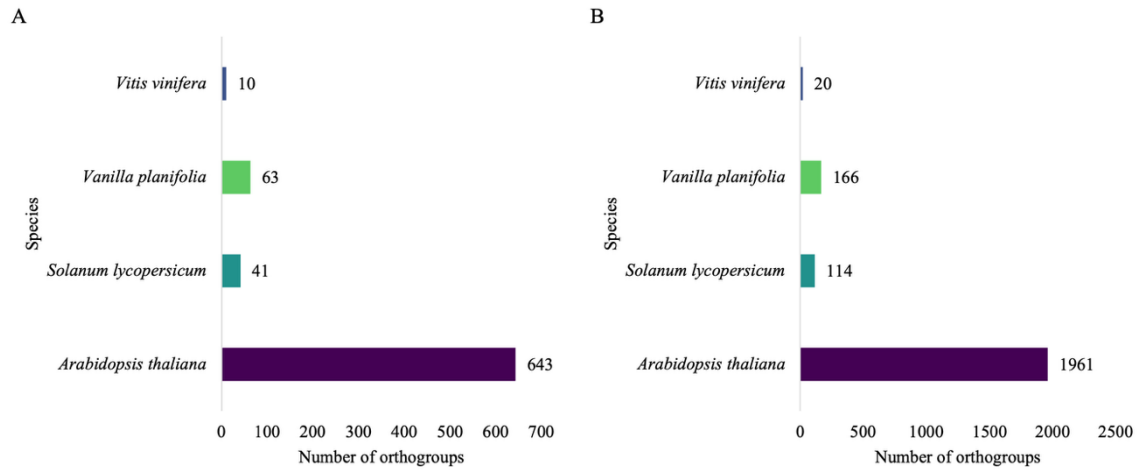


Fig. S2. Number of orthogroups and species-specific genes identified by orthology analysis. (A) Number of orthogroups in *Vitis vinifera*, *Vanilla planifolia*, *Solanum lycopersicum*, and *Arabidopsis thaliana*. (B) Number of species-specific genes associated with each species.

Supplementary Figure S3.

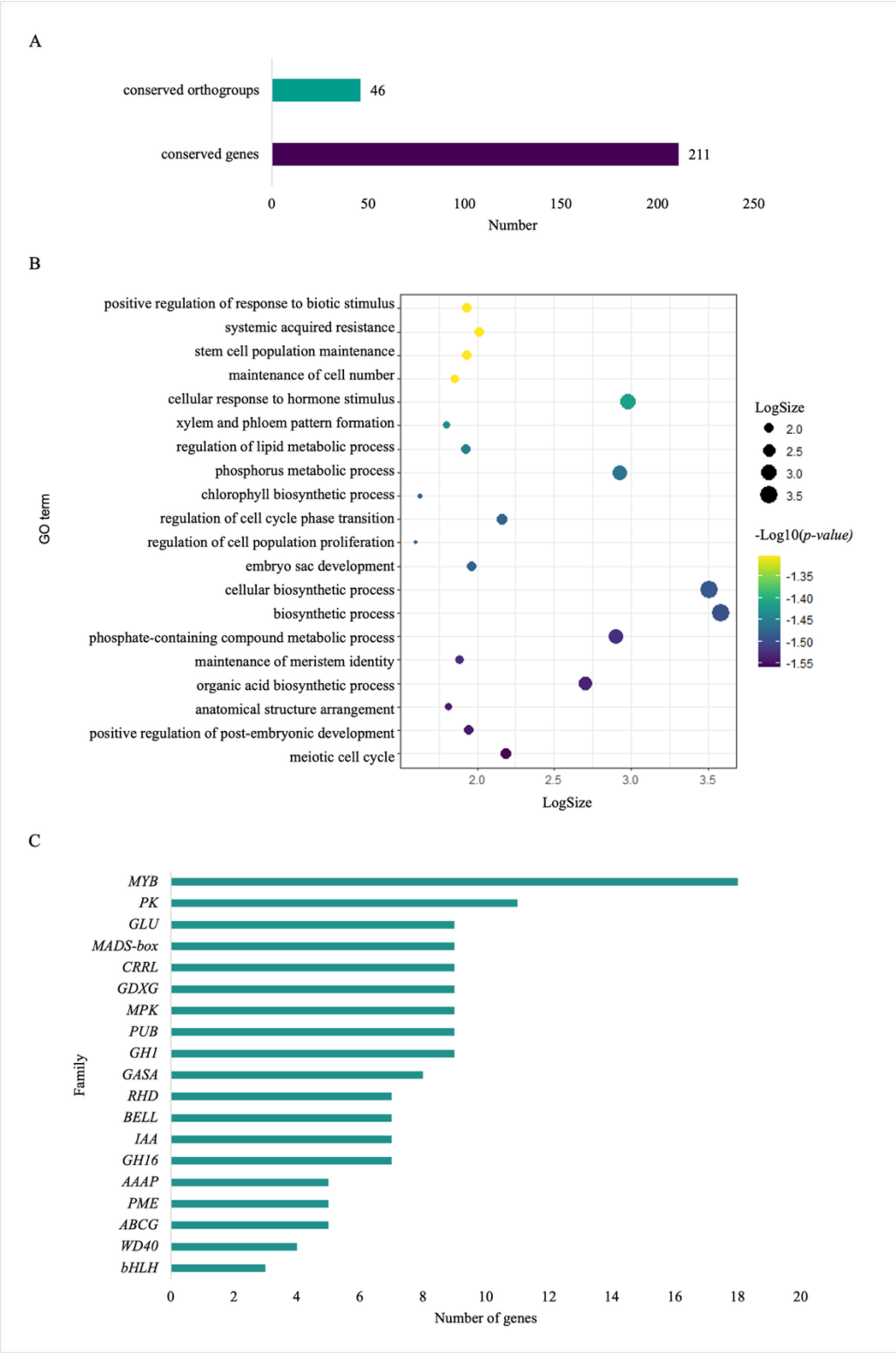


Fig. S3. Conserved orthologs during floral development and the flower-to-fruit transition. (A) Number of orthogroups and genes identified as conserved. (B) Scatter plot of functional enrichment in Planteome showing the 20 most significantly enriched GO Biological Process terms ($p \leq 0.05$).

The y-axis indicates GO terms and the x-axis shows the number of annotations expressed as $\log_{10}$. The color gradient represents statistical significance, with higher $-\log_{10}$(p-value) in yellow and lower values in purple. (C) Number of conserved orthologs associated with floral development and the flower-to-fruit transition, classified by gene family.

Supplementary Figure S4.

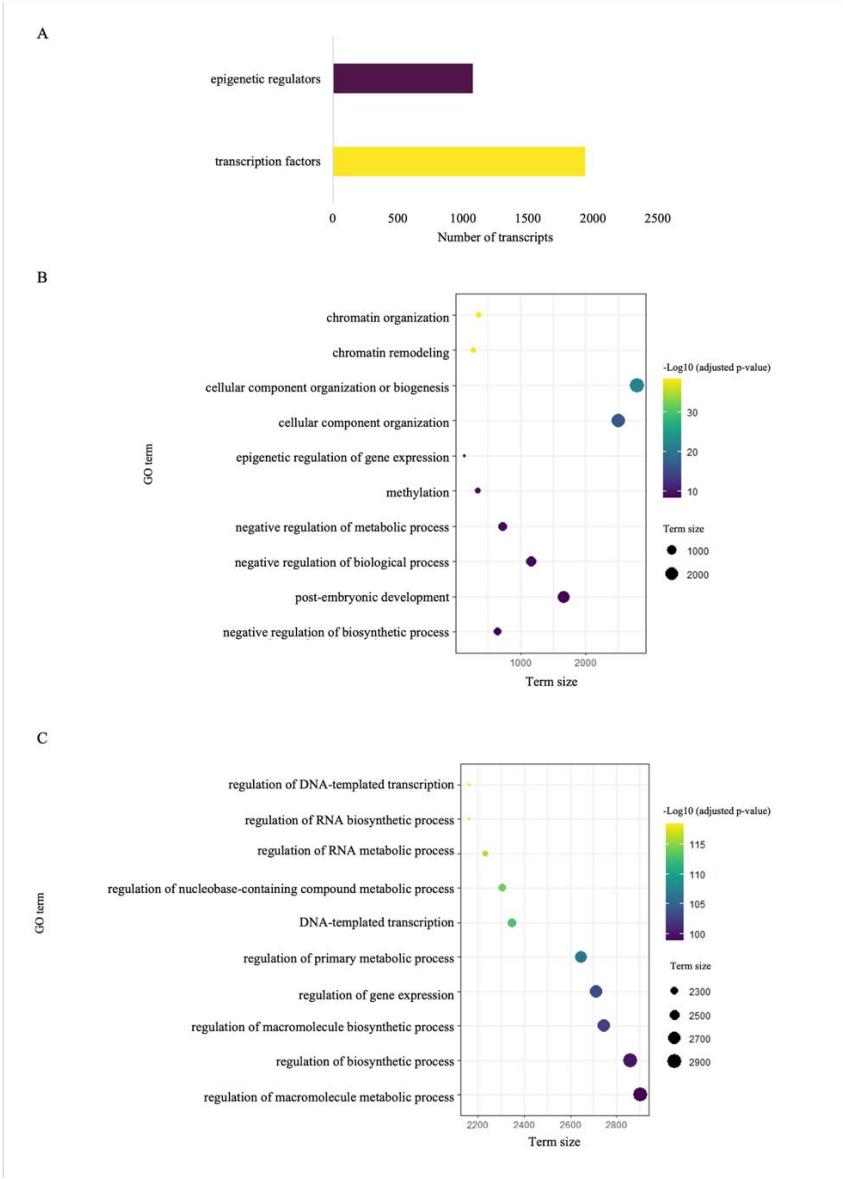


Fig. S4. Epigenetic regulators and transcription factors identified during the flower-to-fruit transition in *V. planifolia*. (A) Number of transcripts annotated as epigenetic regulators or transcription factors. (B) Scatter plot of functional enrichment for epigenetic regulators obtained with gProfiler, showing the ten most significantly enriched GO Biological Process terms ($p \leq 0.05$). The y-axis indicates GO terms and the x-axis shows the number of annotations. The color gradient represents statistical significance, with higher $-\log_{10}$(adjusted p-value) in yellow and lower values in purple. (C) Scatter plot of functional enrichment for transcription factors obtained with gProfiler,

showing the ten most significantly enriched GO Biological Process terms ($p \leq 0.05$). The y-axis indicates GO terms and the x-axis shows the number of annotations. The color gradient represents statistical significance, with higher $-\log_{10}$(adjusted p-value) in yellow and lower values in purple.

Supplementary Figure S5.

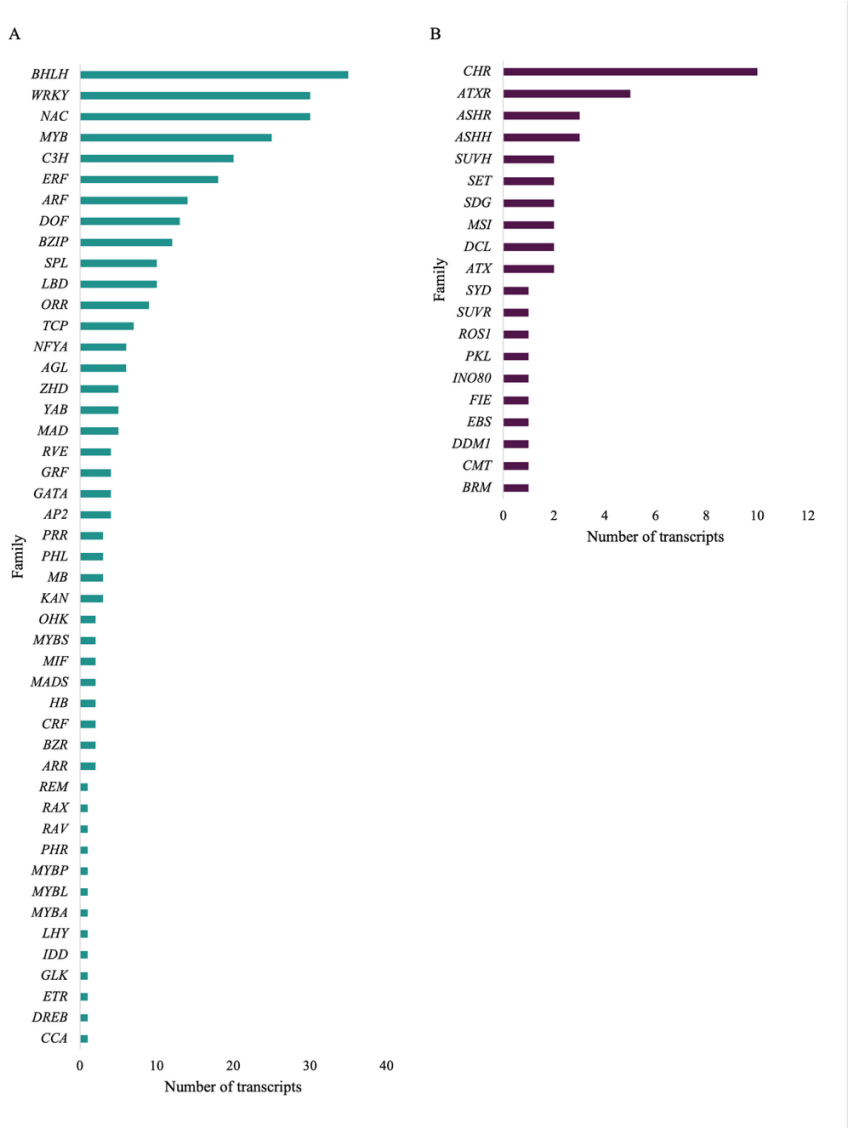


Fig. S5. Gene families represented by transcripts annotated as transcription factors and epigenetic regulators. (A) Transcription factor families. (B) Epigenetic regulator families.

Supplementary Figure S6.

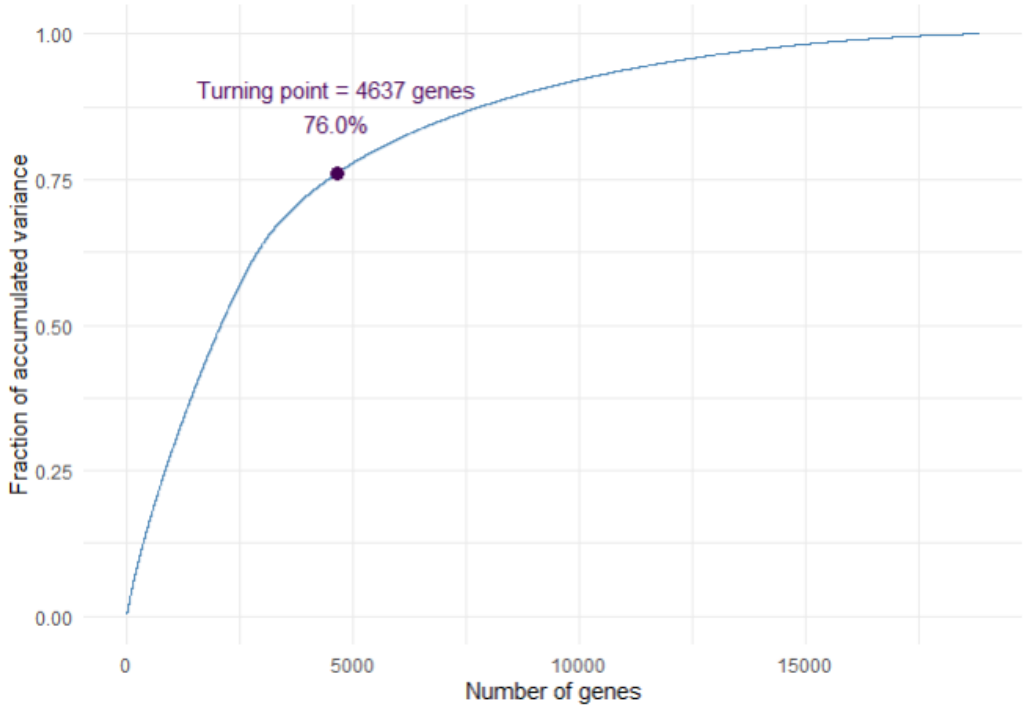
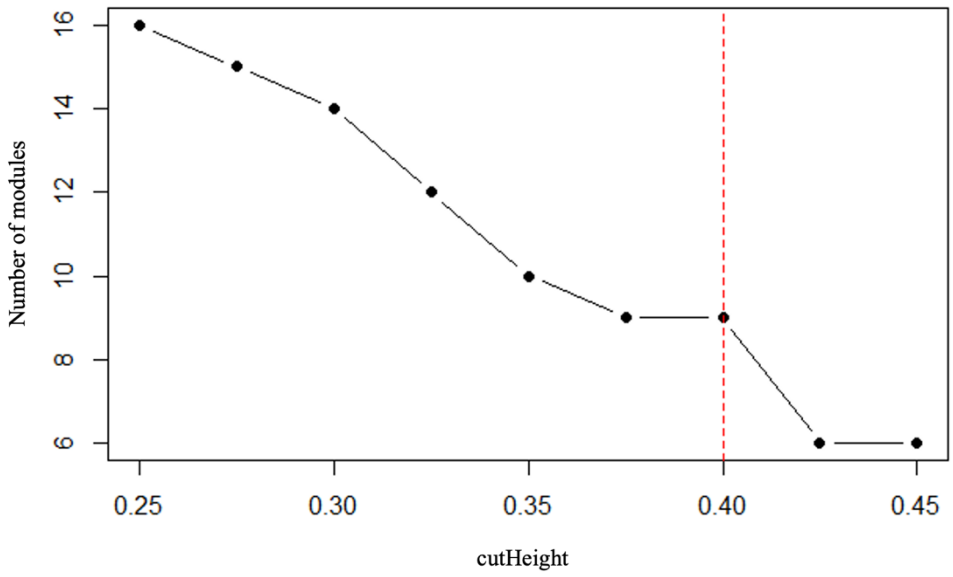


Fig. S6. Cumulative variance curve of genes ranked by variance. The inflection point (4,637 genes) defines the optimal selection threshold, explaining 76% of the total variance of the transcriptome.

Supplementary Figure S7.

A



B

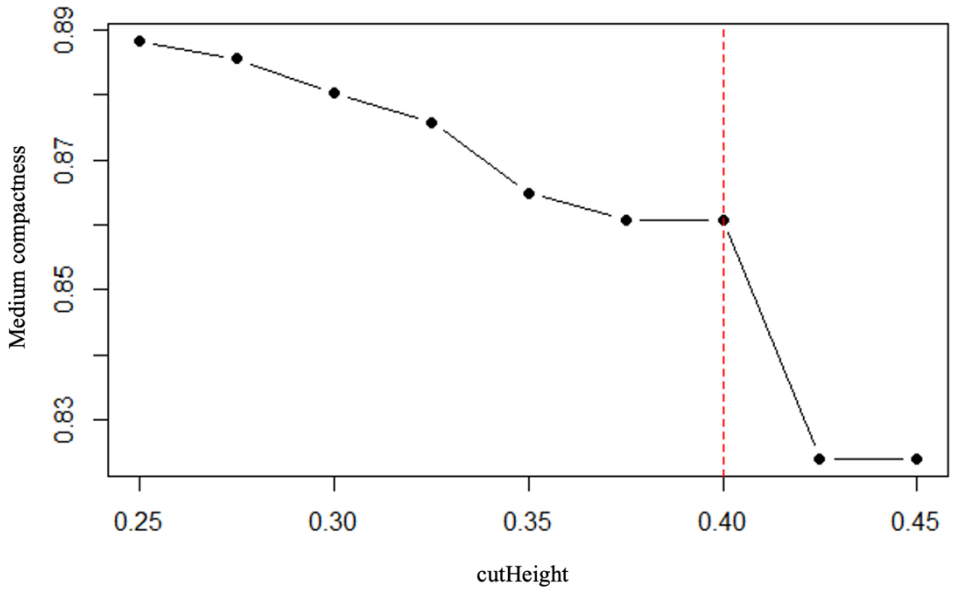


Fig. S7. Optimization of the cutHeight parameter for the *V. planifolia* co-expression network. (A) Number of detected modules as a function of dendrogram cut height. (B) Mean module eigengene connectivity (|kME|) as a function of dendrogram cut height. The value cutHeight = 0.40 (red line) was selected as a compromise, yielding a manageable number of modules (n = 9) while maintaining high intramodular connectivity (>0.85).

Supplementary Figure S8.

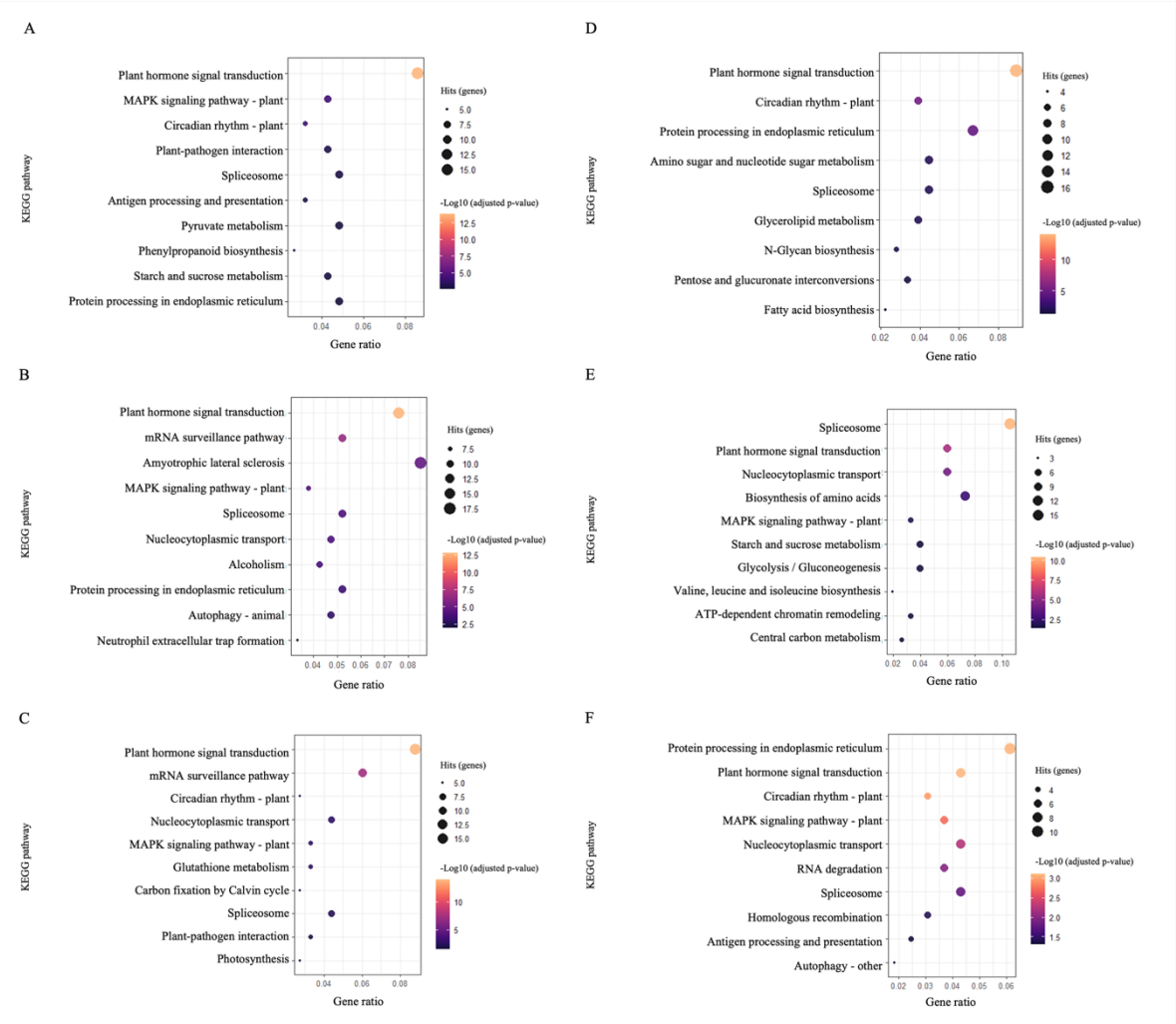


Fig. S8. Functional enrichment of co-expression modules using KEGG pathways. KEGG pathway enrichment analysis was performed using GhostKOALA-annotated sequences against the KEGG database, showing the most significantly enriched metabolic pathways (adjusted p ≤ 0.05) for each module. (A) Green module, (B) dark grey module, (C) brown module, (D) turquoise module, (E) dark orange module, and (F) dark green module. The y-axis indicates KEGG pathways, and the x-axis represents the gene ratio. Dot size reflects the number of genes associated with each pathway, while the color gradient reflects statistical significance, with higher $-\log_{10}$(adjusted p-value) shown in yellow and lower values in purple.

Supplementary Figure S9.

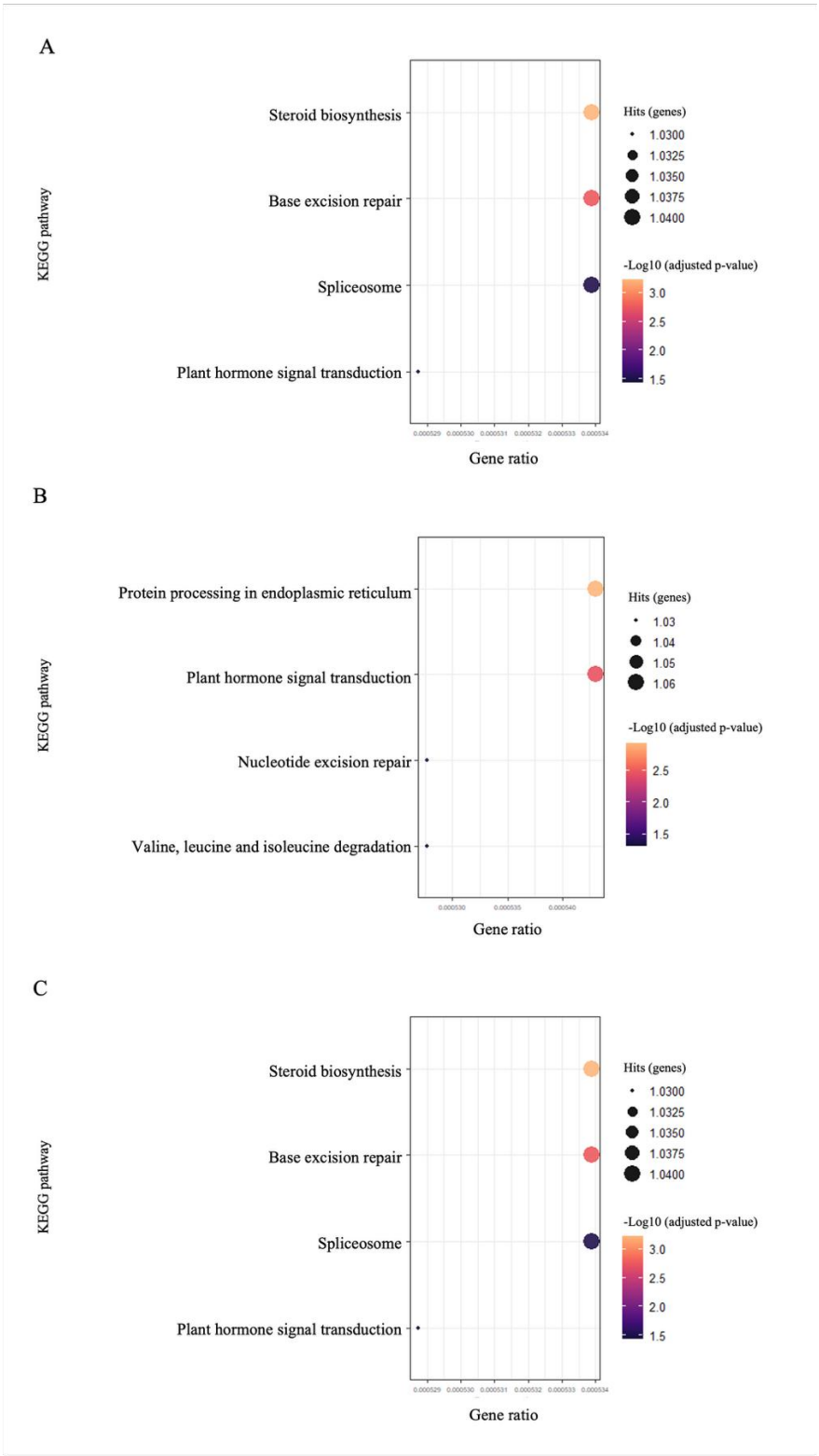


Fig. S9. Functional enrichment of co-expression modules using KEGG pathways. KEGG pathway enrichment analysis was performed using GhostKOALA-annotated sequences against the KEGG database, showing the most significantly enriched metabolic pathways (adjusted p ≤ 0.05) for each module. (A) Red module, (B) pink module, and (C) green-yellow module. The y-

axis indicates KEGG pathways, and the x-axis represents the gene ratio. Dot size reflects the number of genes associated with each pathway, while the color gradient reflects statistical significance, with higher $-\log_{10}$(adjusted p-value) shown in yellow and lower values in purple.

Supplementary Figure S10.

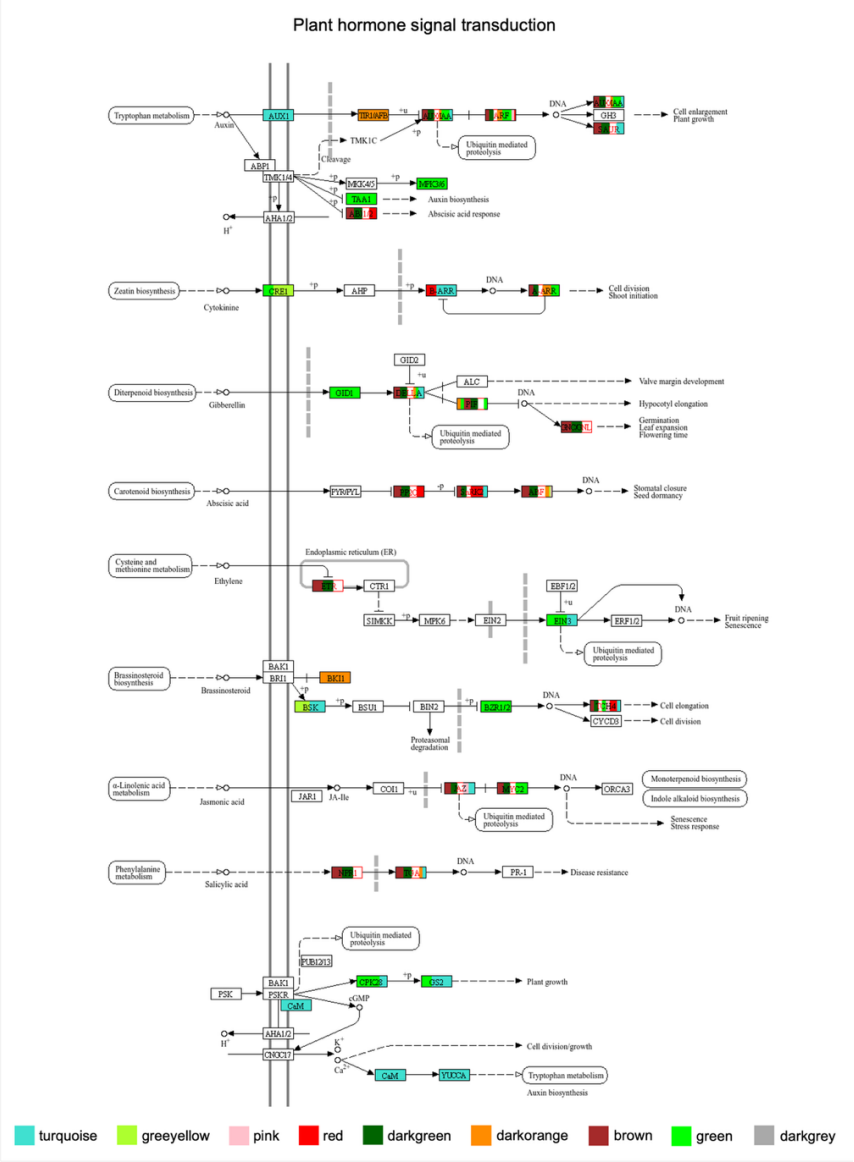


Fig. S10. Co-expression modules mapped onto the plant hormone signal transduction pathway. Each color corresponds to a functional module and highlights the specific steps of the pathway in which its encoded proteins are involved.

Supplementary Figure S11.

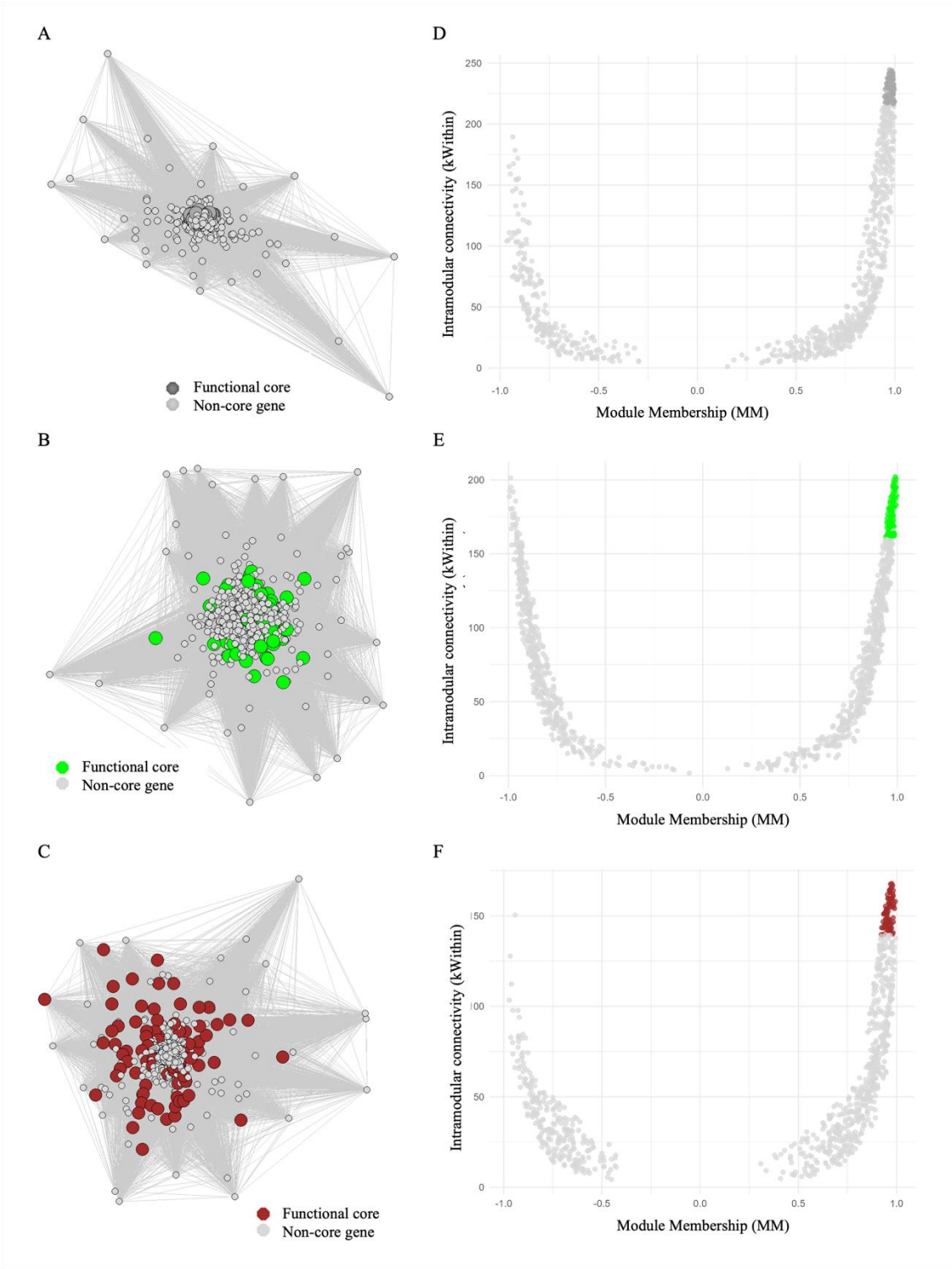


Fig. S11. Transcriptional cores within co-expression modules. (A–C) Intramodular networks of the dark grey, green, and brown modules, highlighting core genes in color. (D–F) Relationships between intramodular connectivity (kWithin) and module membership (MM), showing the high centrality of core genes within each module. Nodes represent genes and edges indicate co-expression interactions; core transcriptional genes are colored according to their corresponding module.

Supplementary Figure S12.

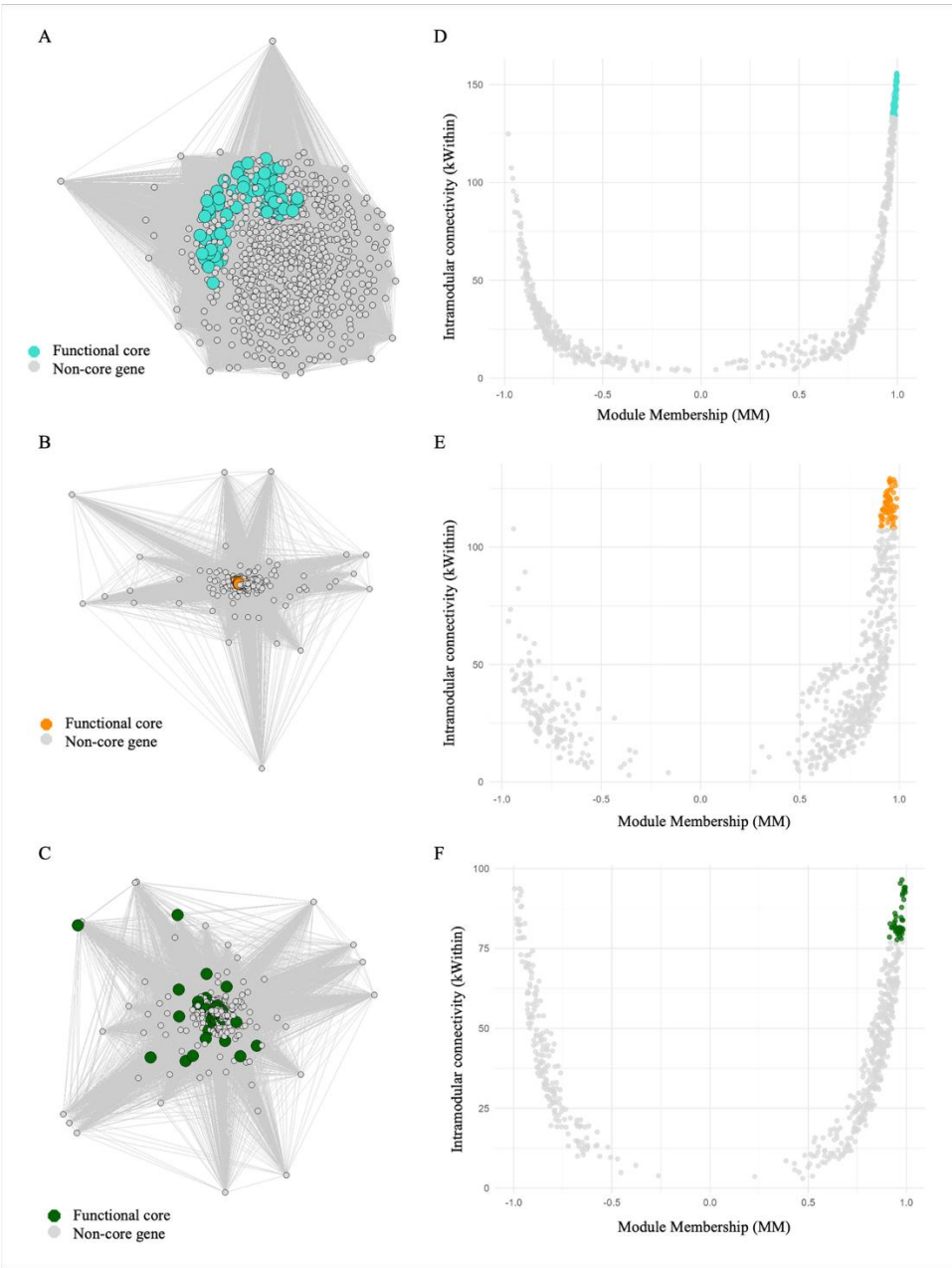


Fig. S12. Transcriptional cores within co-expression modules. (A–C) Intramodular networks of the turquoise, dark orange, and dark green modules, highlighting core genes in color. (D–F) Relationships between intramodular connectivity (kWithin) and module membership (MM), showing the high centrality of core genes within each module. Nodes represent genes and edges indicate co-expression interactions; genes belonging to the transcriptional core are colored according to their corresponding module.

Supplementary Figure S13.

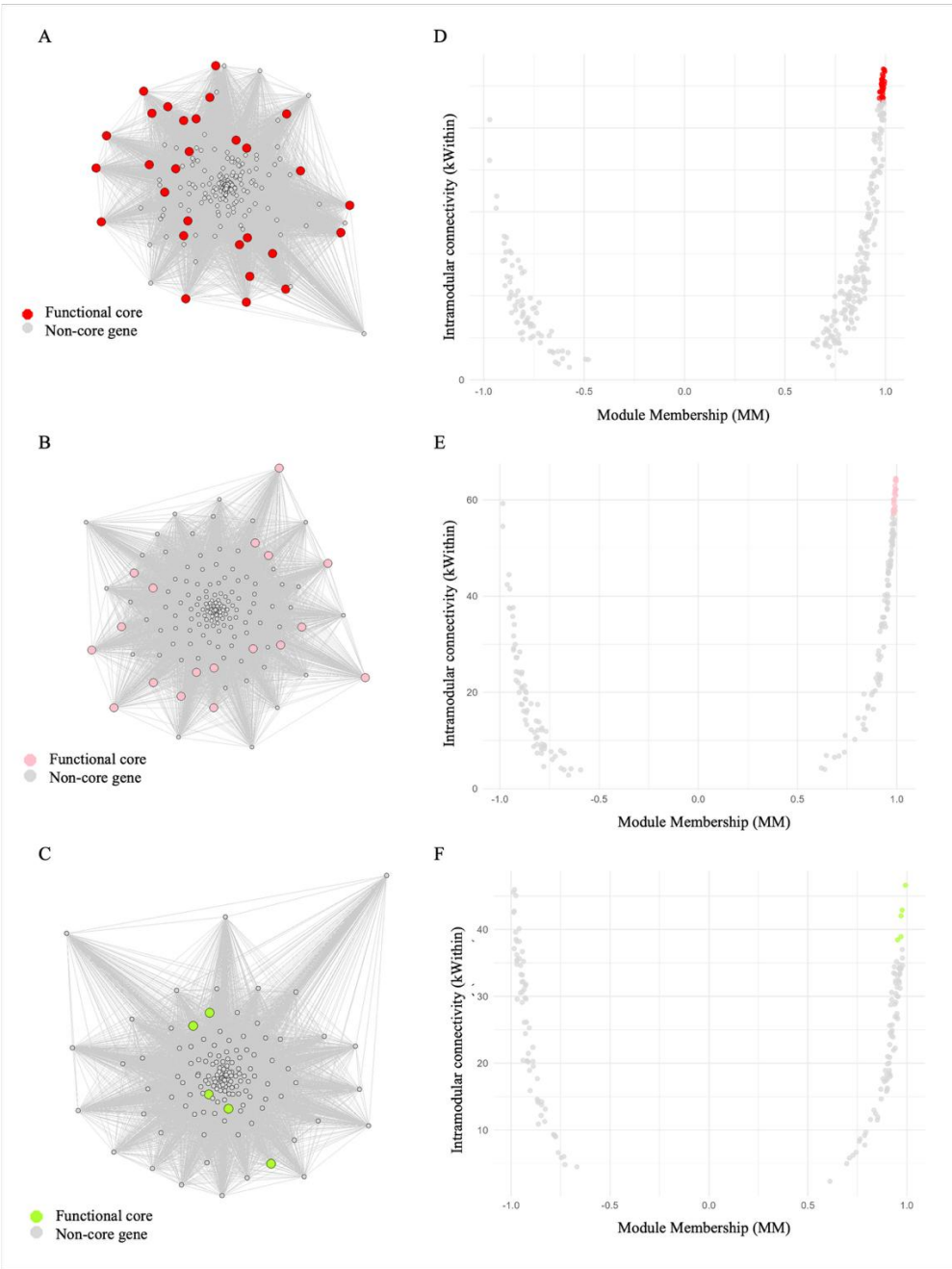


Fig. S13. Transcriptional cores within co-expression modules. (A–C) Intramodular networks of the red, pink, and greenyellow modules, highlighting core genes in color. (D–F) Relationships between intramodular connectivity (kWithin) and module membership (MM), showing the high centrality of core genes within each module. Nodes represent genes and edges indicate co-expression interactions; genes belonging to the transcriptional core are colored according to their corresponding module.

Supplementary Figure S14.

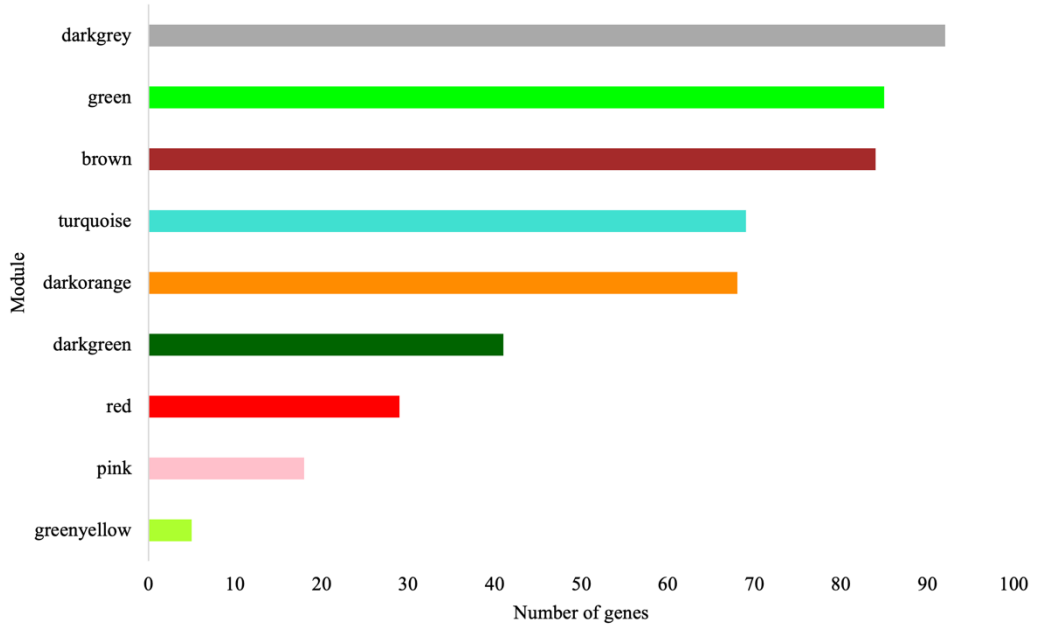


Fig. S14. Distribution of gene numbers per transcriptional core. The bar plot shows the number of genes comprising the transcriptional cores of each co-expression module (darkgrey, green, brown, turquoise, darkorange, darkgreen, red, pink, and greenyellow). Each bar represents the total number of genes assigned to the corresponding core.

Supplementary Figure S15.

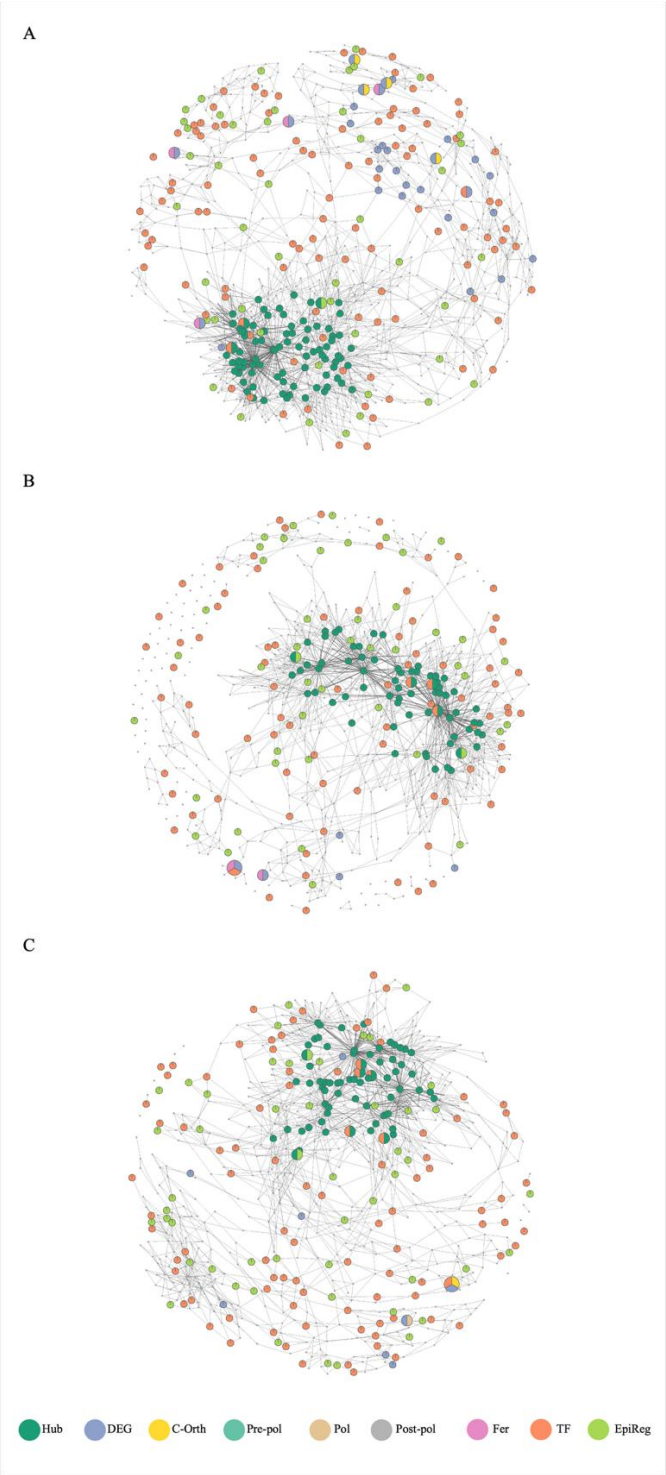


Fig. S15. Co-expression networks of representative transcriptional modules during the flower-to-fruit transition. (A) brown module, (B) turquoise module, and (C) darkorange module. Nodes represent genes and edges indicate co-expression relationships. Node pie charts show the combination of structural and functional labels assigned to each gene. Colors denote labels (Hub, DEG, C-Orth,

Pre-pol, Pol, Post-pol, Fer, TF, and EpiReg), and node size reflects the total number of labels per gene.

Supplementary Figure S16.

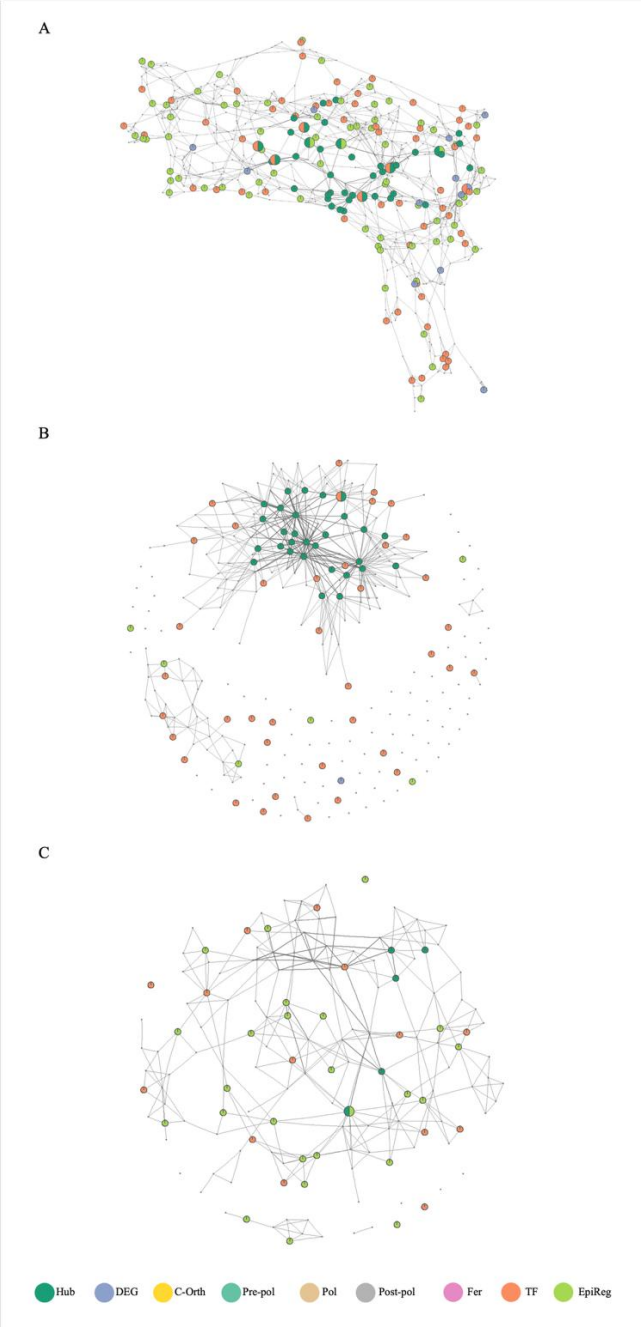


Fig. 16. Co-expression networks of representative transcriptional modules during the flower-to-fruit transition. (A) darkgreen module, (B) red module, and (C) greenyellow module. Nodes represent genes and edges indicate co-expression relationships. Node pie charts show the combination of structural and functional labels assigned to each gene. Colors denote labels (Hub, DEG, C-Orth,

Pre-pol, Pol, Post-pol, Fer, TF, and EpiReg), and node size reflects the total number of labels per gene.